
fabric
FEBRUARY

Thank you to our Fabric February Friends!

bouvet twoday sopr

fabric FEBRUARY

The Ultimate Medallion Load Showdown: Azure v Fabric

Abhi Jayanty & Brian Bønk

Abhi Jayanty

Data Engineer

Quorum

abhi.jayanty@quorum.co.uk

/in/abhinav-j

Brian Bønk

Founder & MVP

ProBI

bbo@probi.dk

/in/brianbonk

brianbonk.dk

probi.dk

Agenda

- Set the scene
- Intro to batch v streaming ingestion
- Battle time!
- Crown the winner?
- Recap and final thoughts

Why are we here?

Why are we here?

- There are multiple leading data platforms for both batch or streaming based ingestion of data
  - Fabric, Azure data stack, Databricks…
- Streaming allows for more than just IoT/telemetry data use cases
- But surely batch is still the way to go?

Quick introduction to batch ingestion

What is batch ingestion?

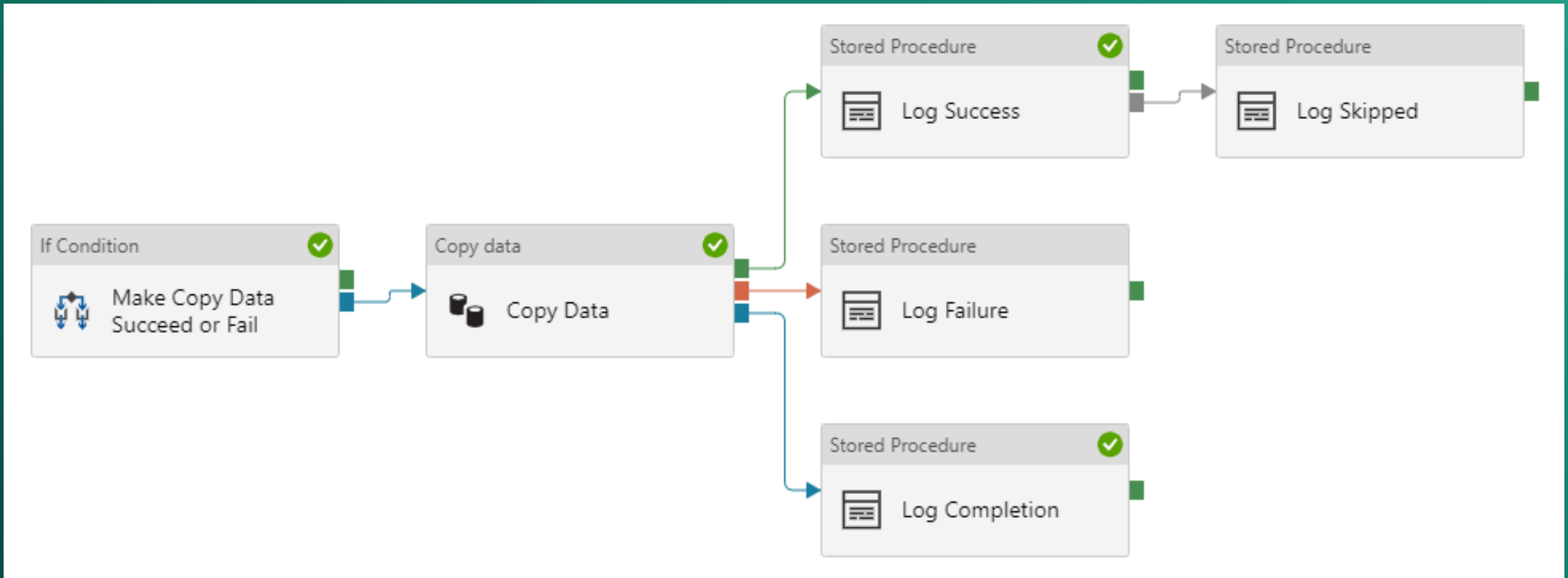
Batch data ingestion is the process of collecting and loading often large volumes of data at scheduled intervals —

hourly, nightly, or on demand

Can be implemented with:

- Azure Data Factory (or Fabric pipelines…)
- Spark notebooks (schedule-based)

What is batch ingestion?



Quick introduction to streaming ingestion

What is streaming ingestion?

Streaming data ingestion captures and processes data continuously as events happen –

in seconds or even milliseconds

Can be implemented with:

- Fabric Real-Time Intelligence
- Azure Event Hubs/Stream Analytics

Fabric Real-Time Intelligence

Enterprise real-time data platforms

Azure Event Hubs

Azure Event Grid

Azure Stream Analytics

Azure Data Explorer

+



Self-service reporting and activation

OneLake

Data Activator

Power BI

=

Real-Time Intelligence in Microsoft Fabric

Fully integrated

No/low-code real-time SaaS

Data platform

Fabric Real-Time Intelligence

Eventstream

Native support for Azure Event/IoT Hubs and various 3rd party sources

No-code experience to transform and clean data in motion

Eventhouse

Logical collection of KQL databases

KQL DB ***KQL DB*** ***KQL DB***

Optimised for time-series data

Real-Time Dashboards

Modular dashboards populated by 'tiles' powered by KQL queries

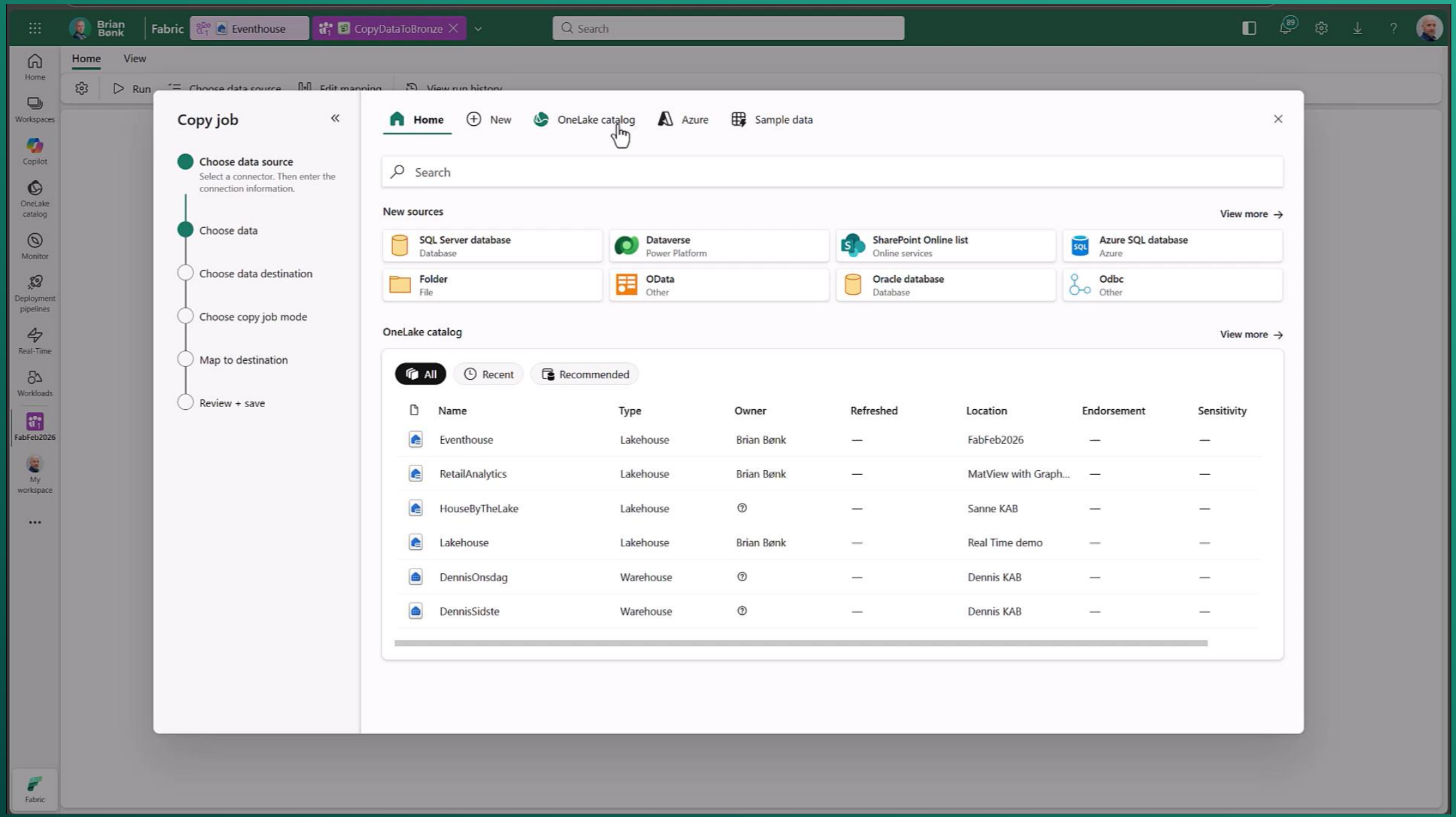
Continuous automatic refresh

#FabricFebruary

Time to battle!

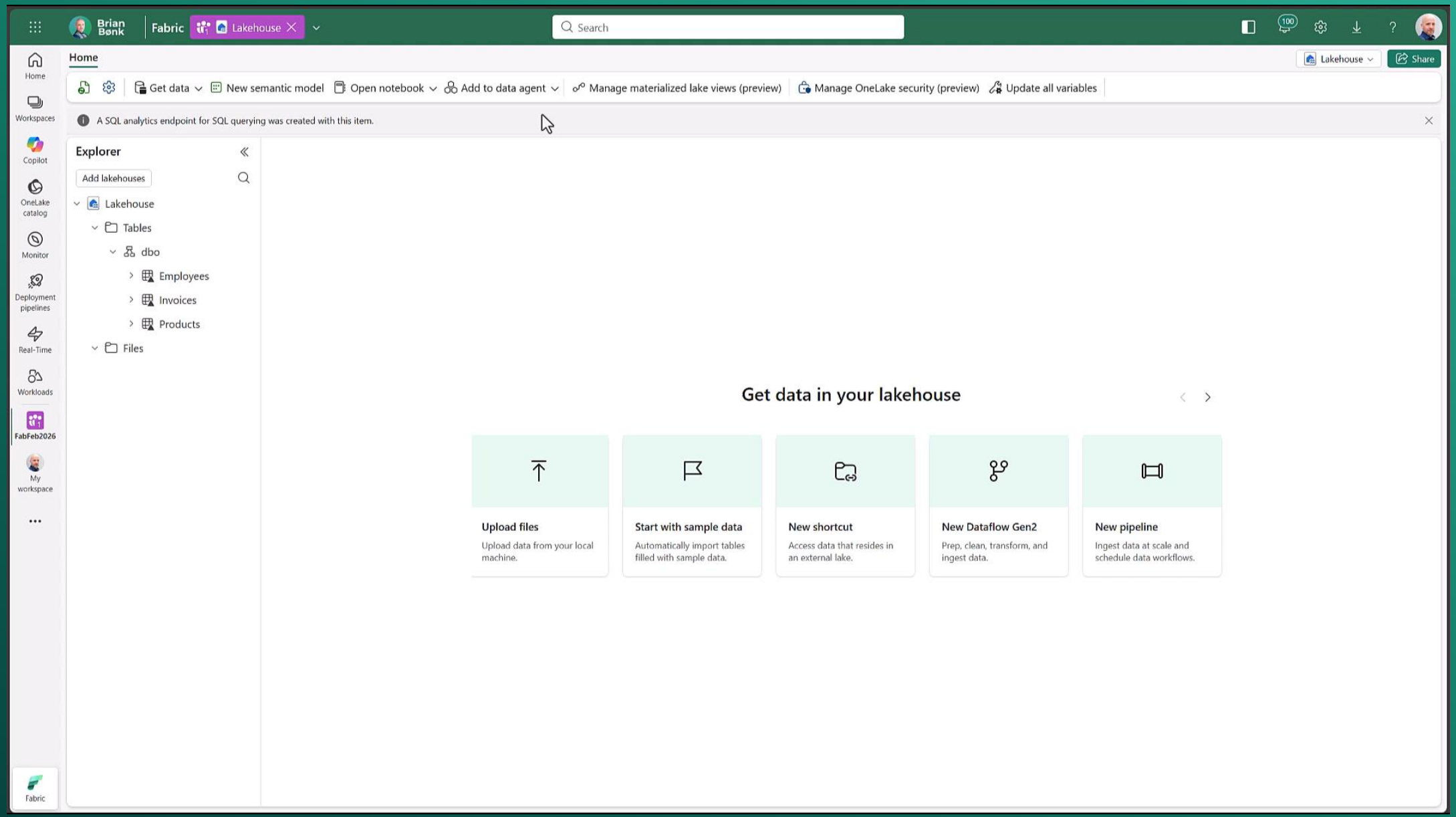
The Rules

- Same dataset
- Same ‘medallion’ enrichment
- Visualize the data at the end:
  - Power BI for batch
  - Fabric Real-Time Dashboards for streaming



#FabricFebruary

https://tabulareditor.com/blog/reintroducing-the-spaceparts-dataset



#FabricFebruary

So who wins?

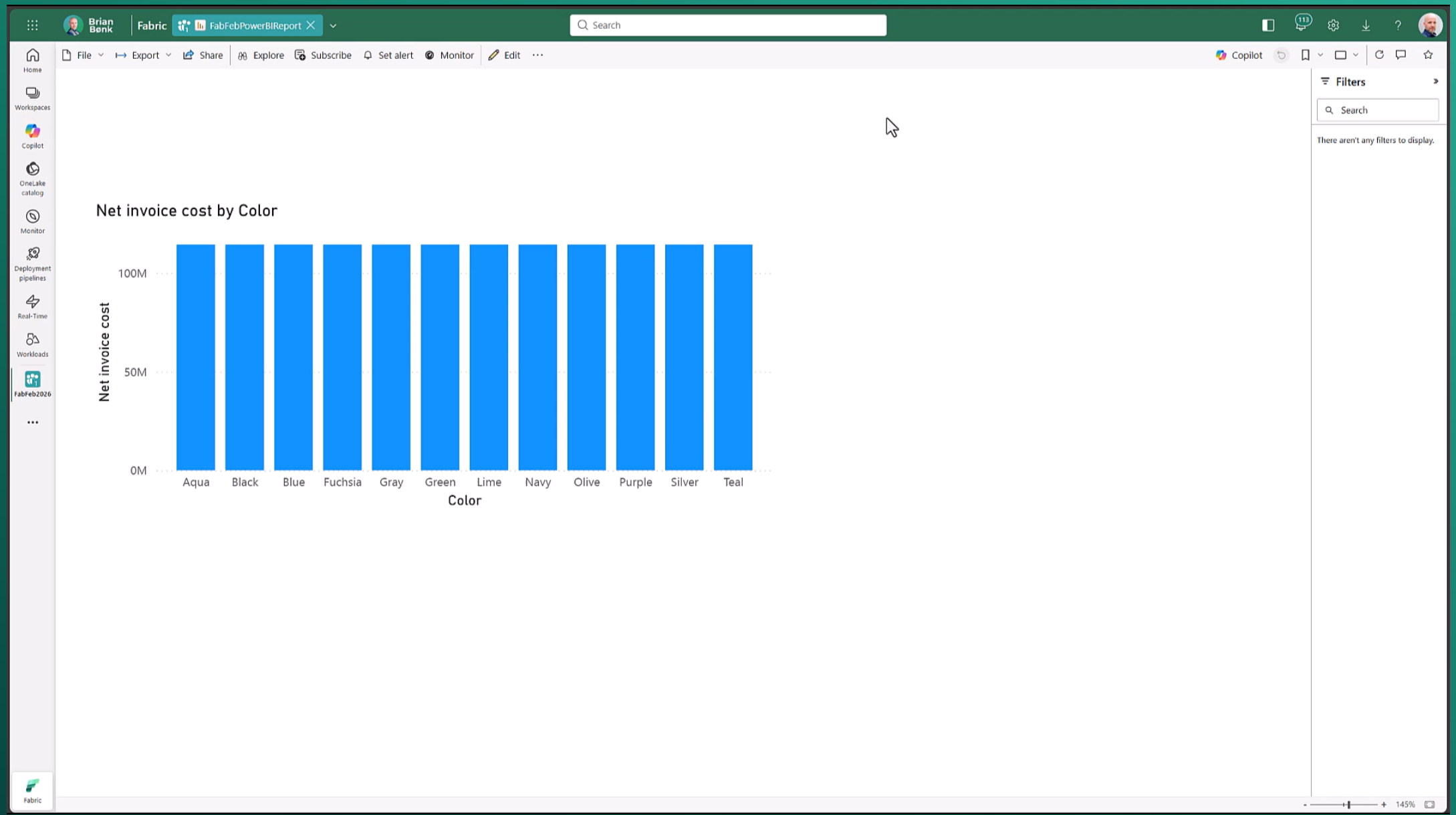
	BATCH	STREAMING
Overall cost		
Speed of ingestion		
Consistency of batches and transactions		
Downstream triggers and actions		

Batch – Advantages

- Predictable
- Better for consistent, transactional data
- Can safely ingest batches of a large volume
- Cheap to run, especially in PaaS world

Batch – Considerations

- Need to create all connections:
  - Source to pipeline
  - Sub-pipeline to the next
  - Pipeline to destination
  - Destination to analysis or visualization layer
- Scheduling — if one pipeline fails or data lands late, then an entire re-run process is required
- Downstream triggers and refreshes not possible (unless you use PowerAutomate or API calls)



#FabricFebruary

Streaming – Advantages

- Super-fast ingestion, data lands whenever it is available, no schedule needed
- Automatic downstream triggers and report refreshes (Fabric Activator)
- SaaS with RTI — time to development is quicker, all transformations can sit in ES or EH (no sub-pipeline mess)

Streaming – Considerations

- Cost — constant background processing can hurt capacity
- Requires better understanding of event-driven data
  - high volume, high duplication, inconsistent data = stronger cleansing
- Often could be ‘overkill’ for a dataset that does not constantly change — use case and scenario is very important

But wait!

We can still be friends ☺

Why not both in your data estate?

- RTI can still use Lakehouse as a destination in Fabric, so essentially becomes another option for ingestion alongside pipelines and notebooks

- ADF or other ETL tools can be very useful for orchestrating and batching out bulk data into messages for event-based destinations (service bus etc.)

Links and Resources

- https://brianbonk.dk/blog/worlds-fastest-medallion-architecture-load/
- Azure Data Explorer — dataexplorer.azure.com
- Kusto Detective Agency
- KQL Reference guide — kql.how

Thank you!

Feedback

fabfeb.app/feedback

fabric FEBRUARY